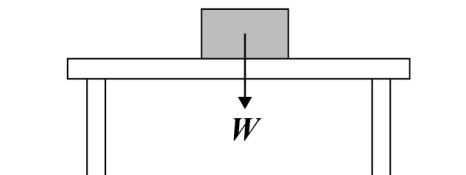


Figure 1 shows an object of weight W on a table.

Figure 1



A student wrongly suggests that W and the normal reaction at the table form an action–reaction pair according to Newton’s third law.

0 2

1

State which force actually forms the action–reaction pair to W .

[1 mark]

0 2

2

Justify your answer.

[1 mark]

Don't write

Which is a statement of Newton's Third Law of Motion?

[1 mark]

A Every force on an object is balanced by an equal and opposite force on the same object.

☐

B When a force acts on an object there is always a second force also acting on it in the opposite direction, but they do not balance.

☐

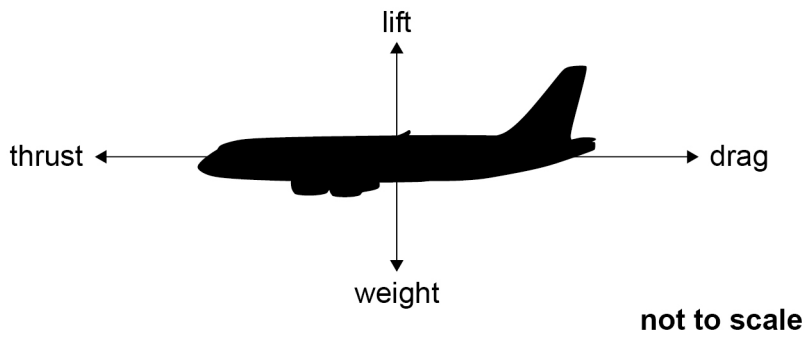
C Whenever an object **X** moves forwards, another object **Y** in that system must move backwards with equal and opposite velocity.

☐

D When object **X** exerts a force on object **Y**, object **Y** must exert an equal and opposite force on object **X**.

☐

The diagram shows an aircraft that is descending at a steady speed and decelerating horizontally.



Which row shows the relationships between lift and weight and between thrust and drag?
[1 mark]

	Relationship between lift and weight	Relationship between thrust and drag	
A	lift < weight	thrust < drag	☐
B	lift < weight	thrust = drag	☐
C	lift = weight	thrust < drag	☐
D	lift = weight	thrust = drag	☐

Turn over for the next question



Do not write

1**2**

Which is in equilibrium?

[1 mark]

A a car moving round a bend at constant speed

☐

B a rocket moving vertically with constant acceleration

☐

C a ball falling vertically at terminal speed

☐

D a train moving horizontally with constant acceleration

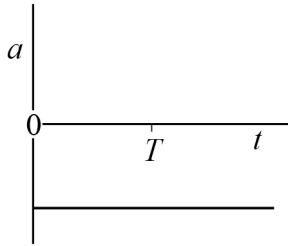
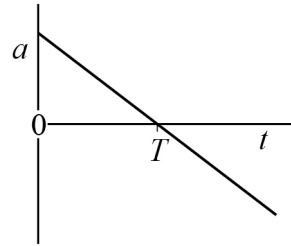
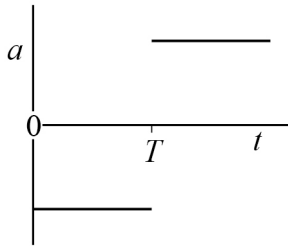
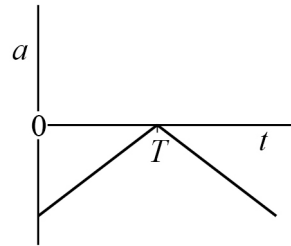
☐

Don't write

1 5

A ball is thrown vertically upwards. It reaches its maximum height at time T and then falls. Air resistance is negligible.

Which graph shows the variation of the acceleration a of the ball with time t ?

[1 mark]**A****B****C****D****A**☐**B**☐**C**☐**D**☐

Don't write

1**7**

A ball is dropped from the fifth floor of a building.

Which single change will increase the terminal speed of the ball?

[1 mark]

A increasing the weight of the ball

☐

B increasing the radius of the ball

☐

C roughening the surface of the ball

☐

D dropping the ball from a higher floor

☐

Which situation does **not** contain an action–reaction pair according to Newton’s third law of motion?

[1 mark]

- A** the weight of a car and the gravitational force on the Earth due to the car
- B** the gravitational force of the Earth on a box and the reaction of the box on the Earth
- C** the frictional force of a tyre on the road and the frictional force of the road on the tyre
- D** the force exerted by an athlete’s foot on a starting block and the reaction of the starting block on the athlete’s foot

☐☐☐☐

Do not write Do not write

0	1
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State the fundamental (base) units for the newton (N).

[1 mark]

fundamental (base) units = _____

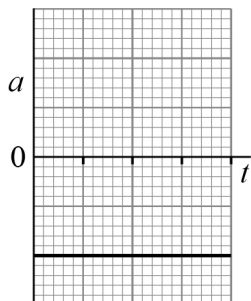
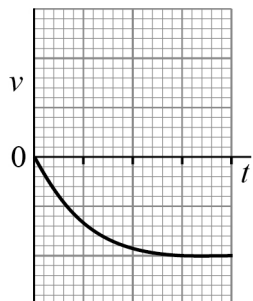
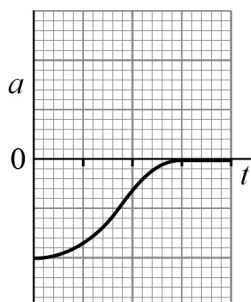
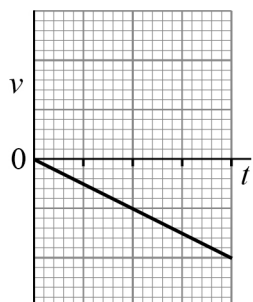
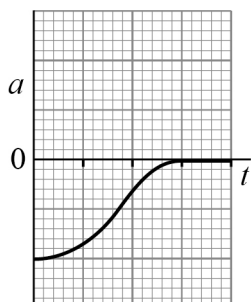
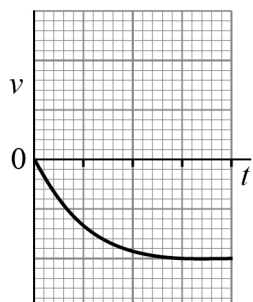
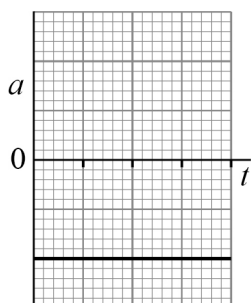
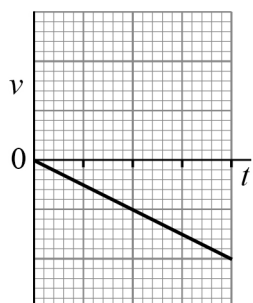
1

1 5

A stone is dropped vertically from rest. It experiences air resistance.

Which pair of graphs shows the variation with time of the velocity v and acceleration a for the motion of the stone?

[1 mark]

A**B****C****D**

Don't write

Cars **P** and **Q** are identical.

P has an initial speed of 13 m s^{-1} . **P** brakes with a constant braking force F .

P travels a distance x before coming to rest.

Q has an initial speed of 15 m s^{-1} . The same constant braking force F is applied to **Q** over the same distance x .

Q does not come to rest but continues at a new constant speed v .

Calculate v .

[3 marks]

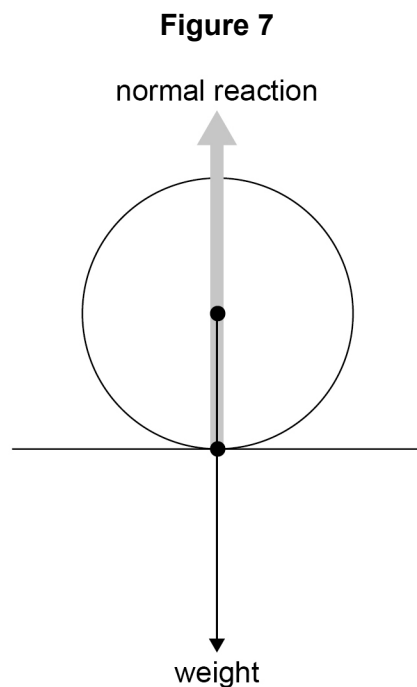
$v =$ _____ m s^{-1}



Do not write

07

Figure 7 shows the two forces acting on the wheel of a stationary racing car that is standing on a horizontal surface.



07.1

The two forces in **Figure 7** are equal in magnitude and act in opposite directions.

Give **one** reason why the two forces are not an action–reaction pair according to Newton’s third law.

[1 mark]



During a race, the racing car enters a straight horizontal section of the track and accelerates. The driving force of the racing car is constant.

0 7 . 2

Explain why the racing car reaches a constant speed.
Refer to one of Newton's laws of motion in your answer.

[3 marks]

0 7 . 3

The racing car is fitted with a system that helps it to overtake another car.

When the system is not in use, as shown in **Figure 8**, the flaps are positioned so that there is no gap between them.

When the system is in use, as shown in **Figure 9**, the flaps rotate so that there is an air gap between them.

Figure 8

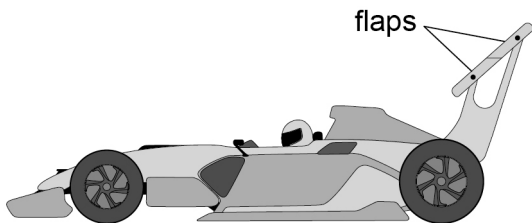
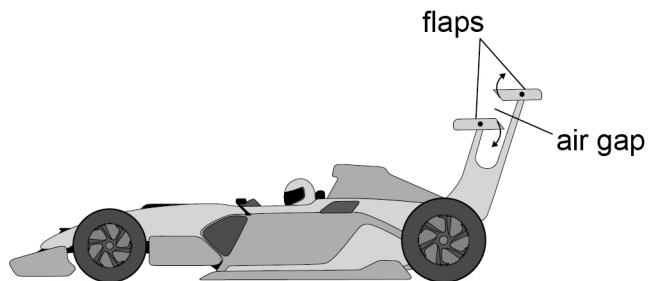


Figure 9



Explain why using the system enables the racing car to overtake the other car.

[2 marks]



Do not write

1 3

A rocket of mass m moves upwards with an acceleration a . It experiences a drag force D .

The thrust of the rocket is:

[1 mark]

A $m(a - g) - D$

☐

B $m(a + g) - D$

☐

C $m(a - g) + D$

☐

D $m(a + g) + D$

☐

Turn over for the next question

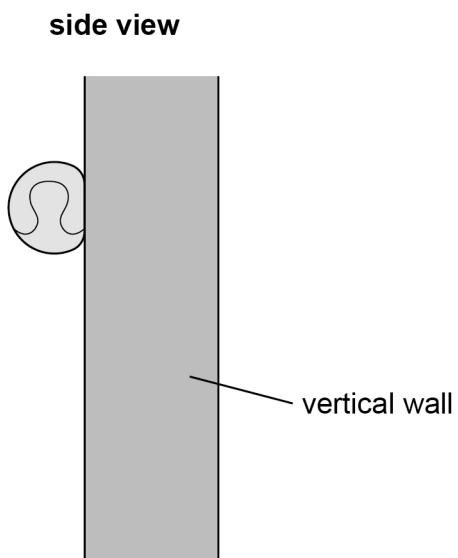
Turn over ►



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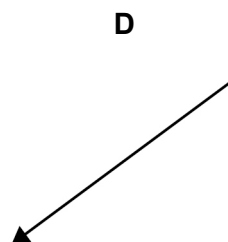
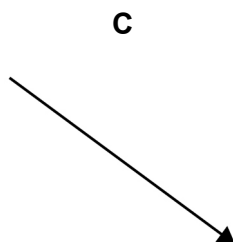
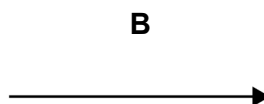
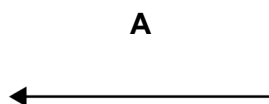
1 4

A ball is travelling horizontally when it hits a vertical wall. The ball compresses when it hits the wall.



What is the direction of the resultant force acting on the ball when the compression is at a maximum?

[1 mark]



Don't write

Figure 2 shows an aircraft of constant weight W accelerating along a horizontal runway before take-off.

Figure 3 shows the aircraft accelerating and climbing after take-off.

Figure 2

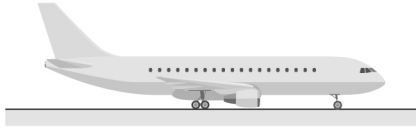
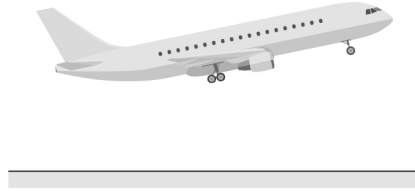


Figure 3



Three of the forces acting on the aircraft are:

L , the lift generated by the aircraft's wings

C , the contact force between the aircraft and the ground

D , the drag experienced by the aircraft.

Explain any changes in L , C and D that happen as the aircraft accelerates along the runway and takes off.

[3 marks]

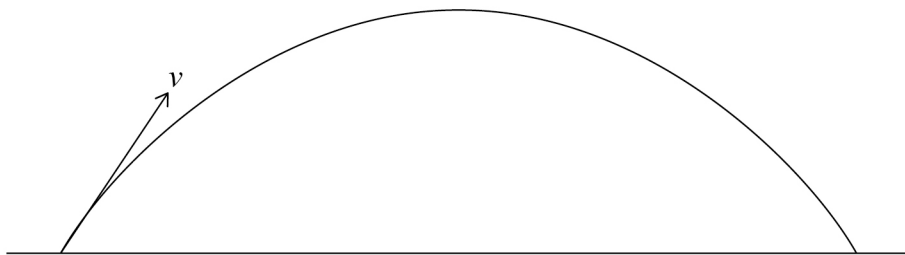
L _____

C _____

D _____

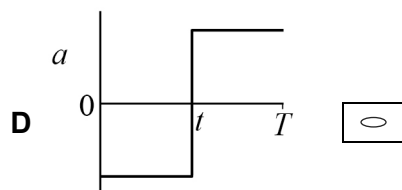
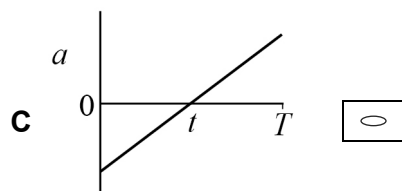
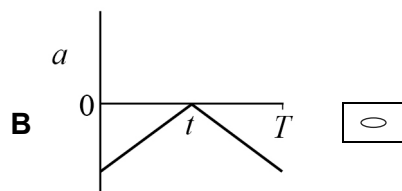
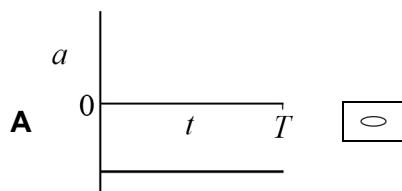
Don't write

An object is projected into the air at time $t = 0$ as shown. It lands when $t = T$. Air resistance is negligible.



Which graph shows the variation with t of the acceleration a of the object?

[1 mark]



Don't write

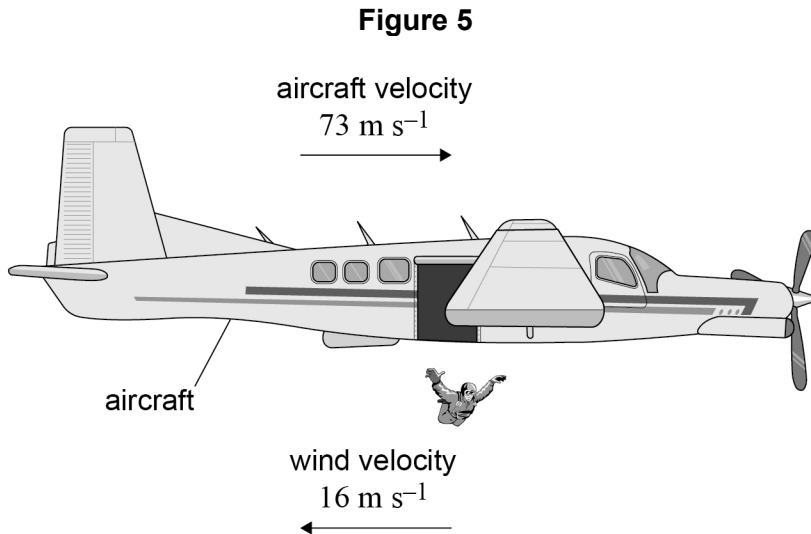
0 6

This question is about a person wearing a parachute.

A parachute is a device that, when open, allows the person to fall slowly through the air.

0 6 . 1

Figure 5 shows the person wearing a closed parachute as she jumps from a moving aircraft.



The aircraft is travelling horizontally with a velocity of 73 m s^{-1} relative to the ground. A wind is moving horizontally in the opposite direction to the motion of the aircraft. The wind has a constant velocity of 16 m s^{-1} relative to the ground.

Sketch, on **Figure 6**, a velocity–time graph for the horizontal component of the person's velocity relative to the ground.

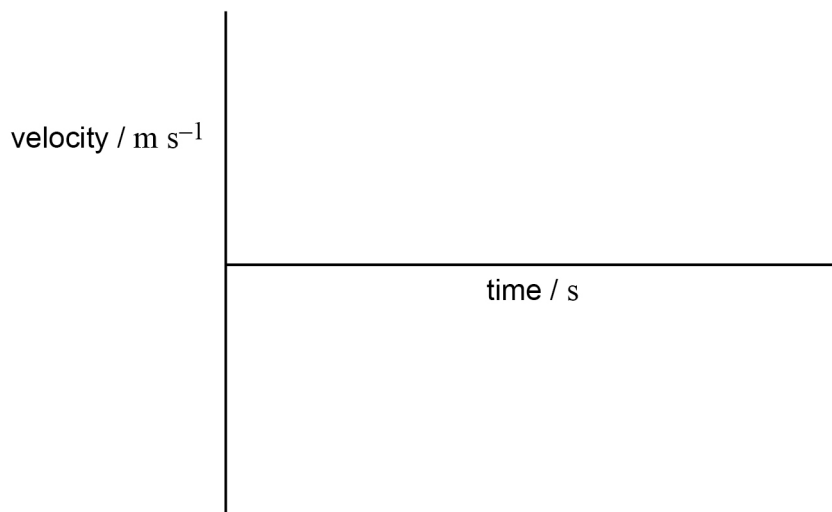
You should:

- label the velocity axis with the values of the initial and final horizontal components of the person's velocity
- continue your graph to the point where the person's horizontal acceleration is zero.

[3 marks]



Figure 6



0 6 . 2

The person opens the parachute as shown in **Figure 7**.

Figure 7



The person then reaches a constant velocity of 18 m s^{-1} at an angle of 29° to the horizontal.

Calculate the time taken for the person to fall a vertical distance of 250 m when at this constant velocity.

[2 marks]

time = _____ s

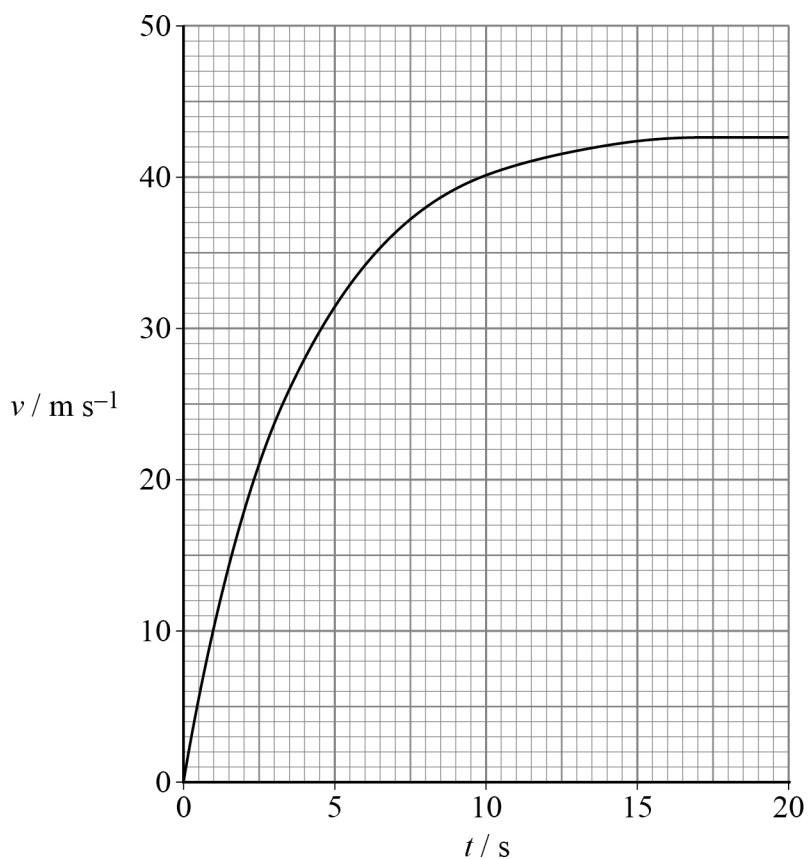
Turn over ►



The person now jumps from the top of a tall building, wearing a parachute.
The parachute is not opened for the first 20 s of the jump.

Figure 8 shows the variation of velocity v with time t for the first 20 s of the jump.

Figure 8



0 6 . 3

The total mass of the person and the parachute is 85 kg.

Calculate the magnitude of the air resistance acting when $t = 5.0$ s.

[4 marks]

air resistance = _____ N



06.

4

Explain the shape of the graph in **Figure 8**.

[4 marks]

Turn over for the next question



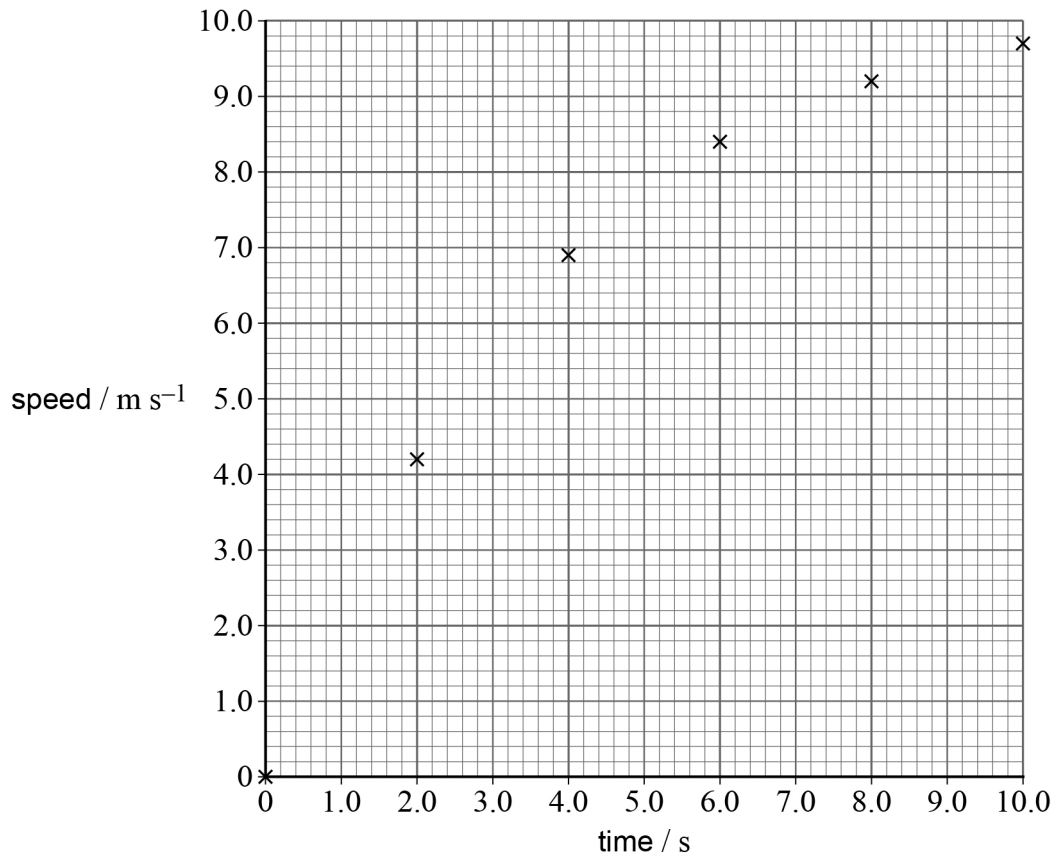
Don't write

1 1

A bike rider accelerates from rest along a straight horizontal road.

Figure 12 shows the variation of the rider's speed with time.

Figure 12



1 1

1

Draw, on **Figure 12**, a line of best fit.

[1 mark]

1 1

2

The total mass of the bike and rider is 95 kg.

Determine, using **Figure 12**, the maximum resultant force on the bike and rider.

[3 marks]

maximum resultant force = _____ N



1 1 . 3 The force exerted by the rider on the bike is constant.

Explain the shape of the graph in **Figure 12**.
No calculations are required.

[3 marks]

7

END OF SECTION B



Section C

Each of the questions in this section is followed by four responses, **A**, **B**, **C** and **D**.

For each question select the best response.

Only **one** answer per question is allowed.

For each question, completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS



If you want to change your answer you must cross out your original answer as shown.



If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown.

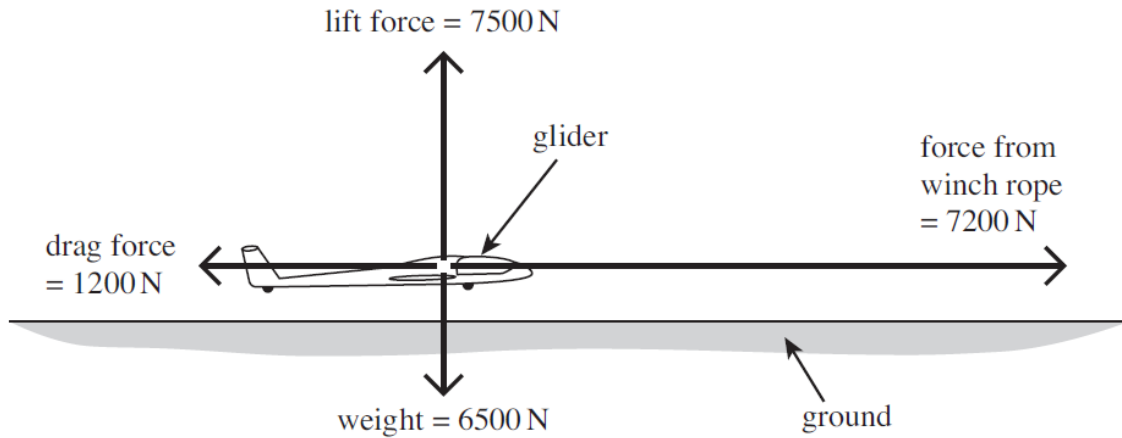


You may do your working in the blank space around each question but this will not be marked.
Do **not** use additional pages for this working.

0 1

Giders can be launched using a machine that pulls a rope attached to the glider. **Figure 1** shows the forces acting on the glider at one instant during the launch.

Figure 1



The combined weight of the glider and pilot is 6500 N.

0 1

. 1

Show that the magnitude of the resultant force acting on the glider is approximately 6100 N.

[2 marks]

0 1

. 2

Determine the angle from the horizontal of the resultant force acting on the glider.

[2 marks]

angle = _____ degrees

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